

</Hacker4Help>



Analysis of Windows OS Threats: PowerShell Abuse, Unauthorized Account Creation, and Log Clearing using Wazuh SIEM

RITIK PATEL

SOC – Threat Research Intern

</Hacker4Help>

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

INDEX

Sr. No	Title	Pg NO
1	Introduction	3
2	Lab Environment & Attack Scenario	4
3	Log Collection & Audit Configuration	6
4	Attack Scenario	15
5	Attack Simulation	17
6	Detection And Analysis Using Wazuh	29
7	Conclusion	32

1. Introduction

In the previous phase of this project, a complete enterprise lab environment was successfully established, including Active Directory Domain Services (AD DS) and a Security Information and Event Management (SIEM) solution using Wazuh for centralized log collection and analysis.

Building upon this foundation, this phase focuses on analyzing common Windows operating system threats by simulating attacker behavior within a controlled lab environment. Unlike external attacks such as brute-force attempts, this scenario assumes that the attacker has already gained initial access to the system and is performing post-exploitation activities.

The simulation includes multiple attacker techniques such as PowerShell abuse, unauthorized account creation, account modification, and defense evasion through log clearing. These activities generate critical Windows event logs, which are then monitored and analyzed using Wazuh SIEM.

The objective of this phase is to understand how such threats appear in system logs, how they can be detected using SIEM tools, and how multiple events can be correlated to identify malicious activity in a real-world Security Operations Center (SOC) environment.

The objectives of this phase are:

- To simulate common Windows operating system threats in a controlled lab environment
- To generate and analyze security logs related to attacker activity
- To detect suspicious behavior using Wazuh SIEM
- To correlate multiple events and understand the attack lifecycle
- To gain practical experience in SOC-based monitoring and investigate

2. Lab Environment & Attack Scenario

2.1 Lab Architecture

The lab environment used for this simulation is the same as the previously established setup, designed to replicate a real-world enterprise network. It consists of multiple virtual machines configured within a VirtualBox environment using a NAT network.

Components of the Lab:

1. Domain Controller: Windows Server 2022 with Active Directory Domain Services (AD DS)
2. Client Machine: Windows 10 and Ubuntu system joined to the domain. Windows 10 serves as the primary endpoint for attack simulation
3. SIEM Platform: Wazuh collects and analyzes logs from endpoints. It is used for monitoring and detection of suspicious activity
4. Network Configuration: All the machines are connected within the same VirtualBox NAT network

2.2 Attack Scenario

In this simulation, it is assumed that the attacker has already gained access to the Windows 10 client system. Instead of focusing on initial access (such as brute-force attacks), this scenario emphasizes post-exploitation activities, which are commonly observed after a system compromise.

The attacker performs a sequence of actions to establish persistence, manipulate system behavior, and evade detection mechanisms. These actions generate various Windows event logs, which are collected and analyzed using Wazuh SIEM.

Simulated Attacker Activities:

</Hacker4Help>

- Execution of suspicious PowerShell commands
- Creation of a new local user account
- Modification of the created user account
- Attempt to alter security configurations (Windows Defender)
- Clearing of Windows Security logs to remove evidence

2.3 Assumptions

The following assumptions are made for this simulation:

- The attacker has already obtained administrative access to the system
- Wazuh agent is successfully installed and forwarding logs
- The environment is controlled and isolated for safe testing

3. Log Collection and Audit Configuration

3.1 Overview

To ensure effective detection of attacker activity, it is essential to both enable detailed logging on the Windows endpoint and configure log forwarding to the Wazuh.

In this section, critical logging features such as PowerShell logging, process creation logging, and account management auditing are enabled. These logs are then forwarded to Wazuh for centralized monitoring and analysis.

3.2 Enabling PowerShell Logging

PowerShell Script Block Logging was enabled to capture executed commands and scripts.

This allows detection of:

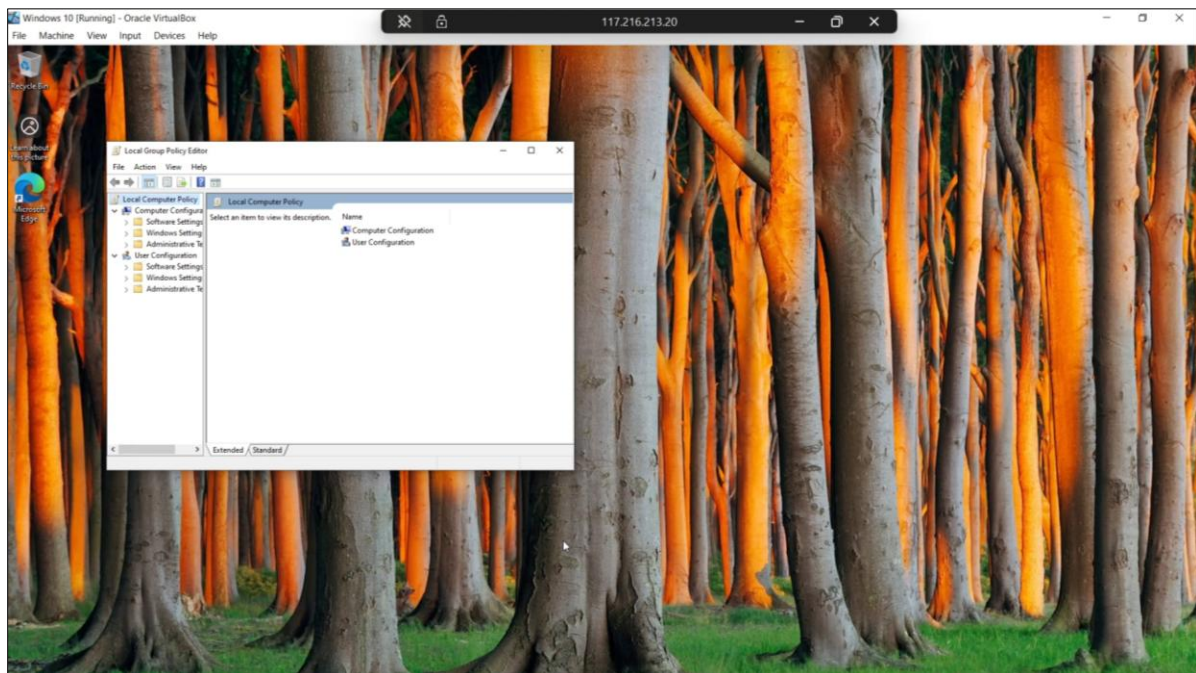
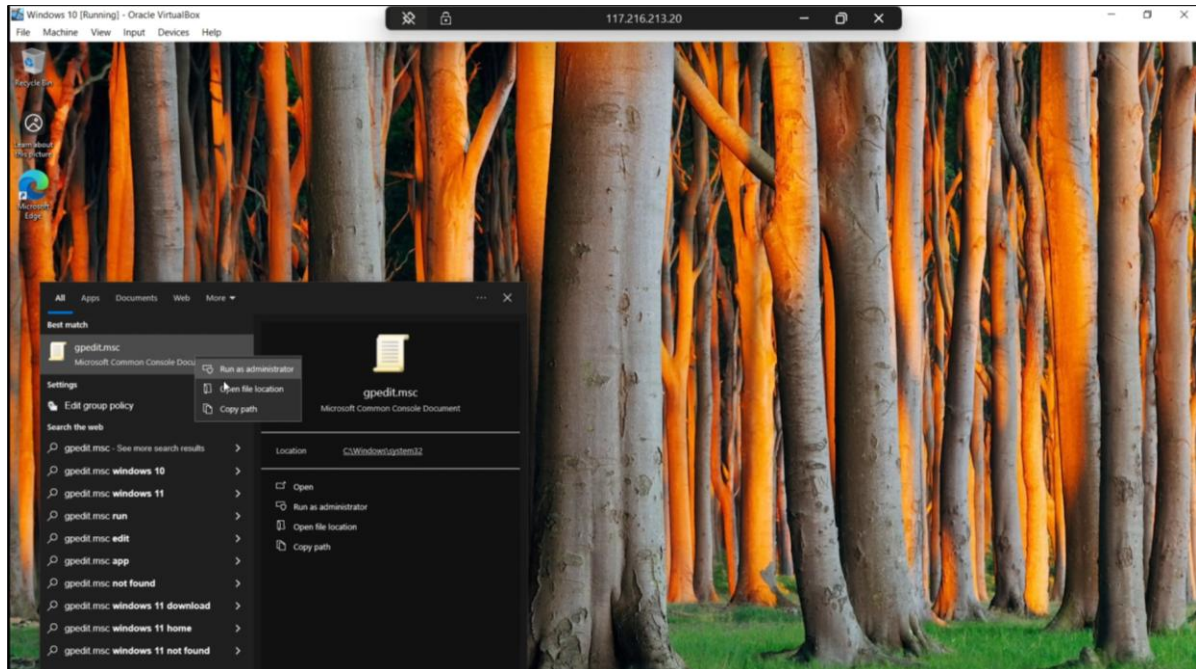
- Suspicious command execution
- Script-based attacks

Key Event ID: 4104 – Script Block Logging

Steps:

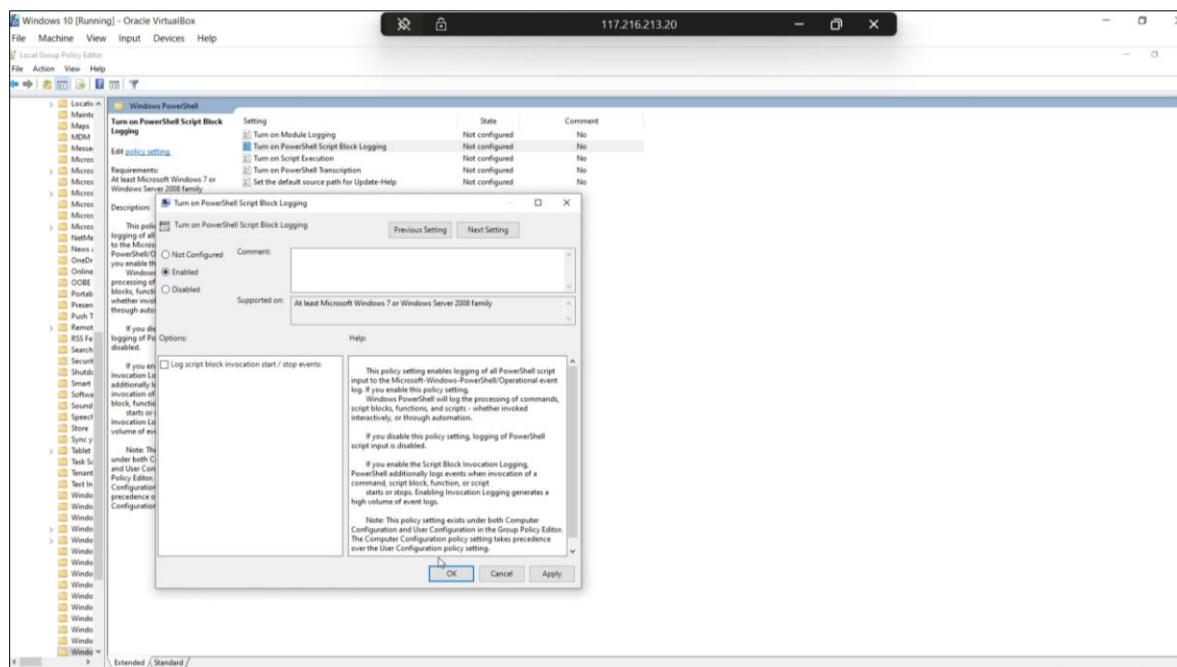
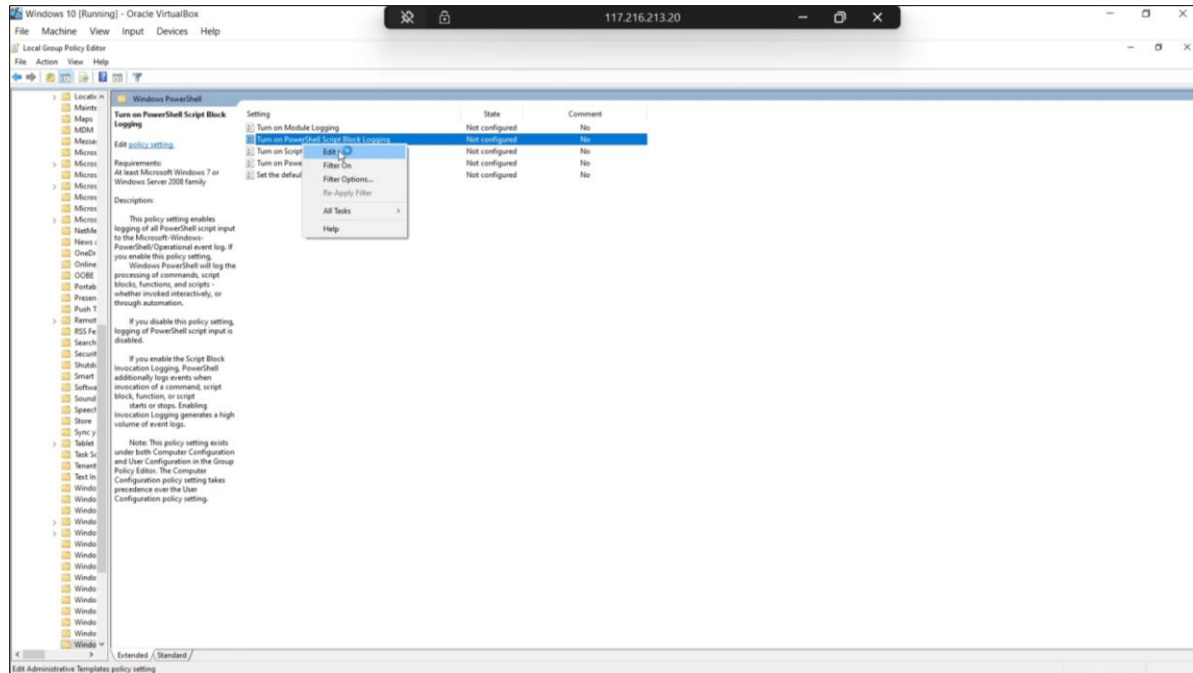
1. Open Group Policy Editor: gpedit.msc
2. Navigate to: Computer Configuration → Administrative Templates → Windows Components → Windows PowerShell
3. Enable: Turn on PowerShell Script Block Logging

</Hacker4Help>



Ritik Patel
SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>



3.3 Enabling Process Creation Logging

Process creation logging was enabled to track executed processes along with command-line arguments.

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>

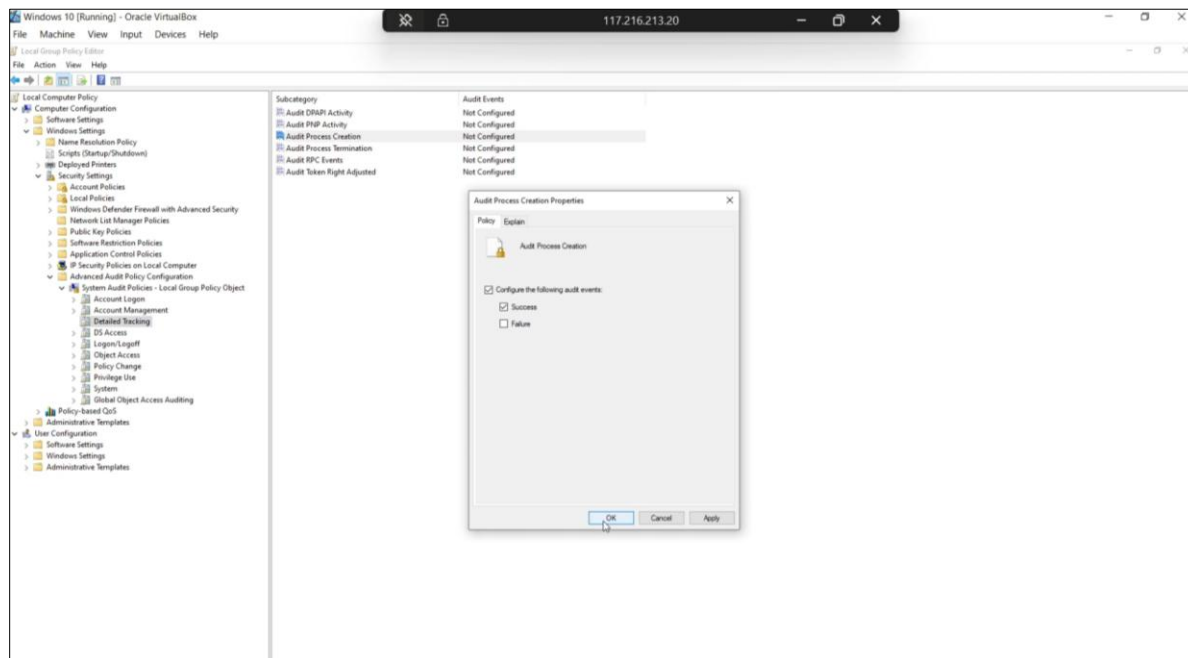
This provides visibility into:

- Executed programs
- Parent-child process relationships

Key Event ID: 4688 – Process Creation

Steps:

1. Open Group Policy Editor: gpedit.msc
2. Navigate to: Computer Configuration → Security Settings → Advanced Audit Policy Configuration → System Audit Policies → Detailed Tracking
3. Enable: Audit Process Creation (Success)



3.4 Enabling Account Management Auditing

Account management auditing was enabled to monitor user-related activities.

This includes:

- User creation
- Account modification
- Privilege-related changes

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

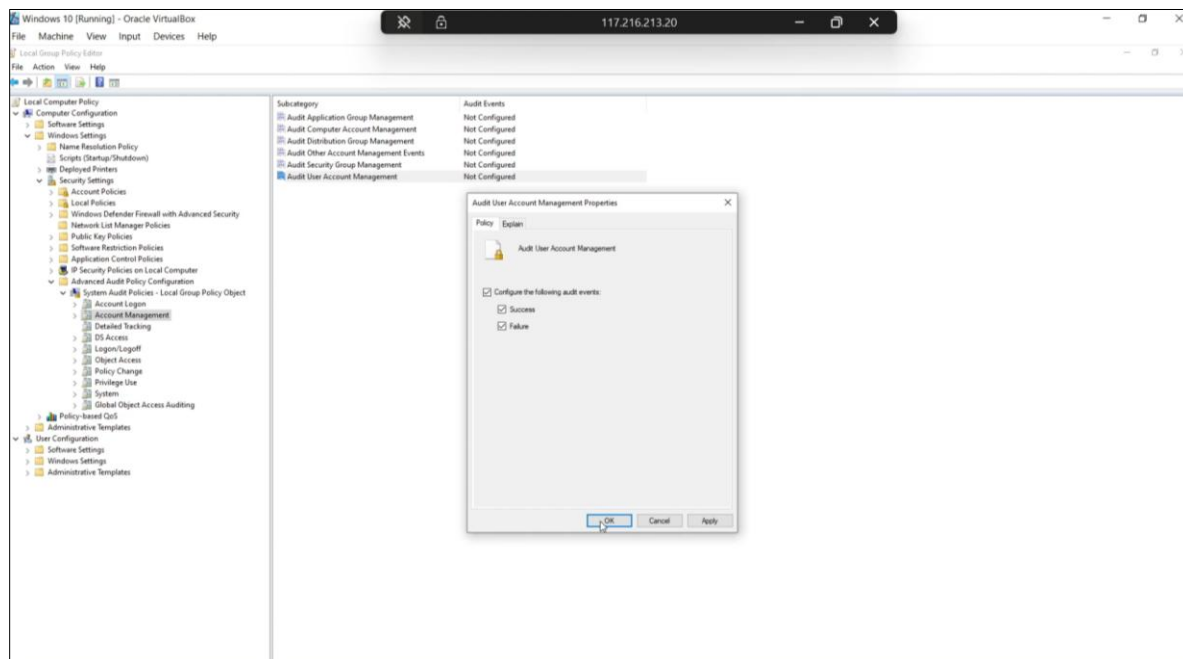
</Hacker4Help>

Key Event IDs:

- 4720 – User Created
- 4738 – User Modified

Steps:

1. Open Group Policy Editor: gpedit.msc
2. Navigate to: Computer Configuration → Security Settings → Advanced Audit Policy Configuration → System Audit Policies → Account Management
3. Enable: Audit User Account Management



3.5 Enabling Command-Line Logging

To enhance visibility, command-line arguments for process creation were enabled using a registry modification.

Steps:

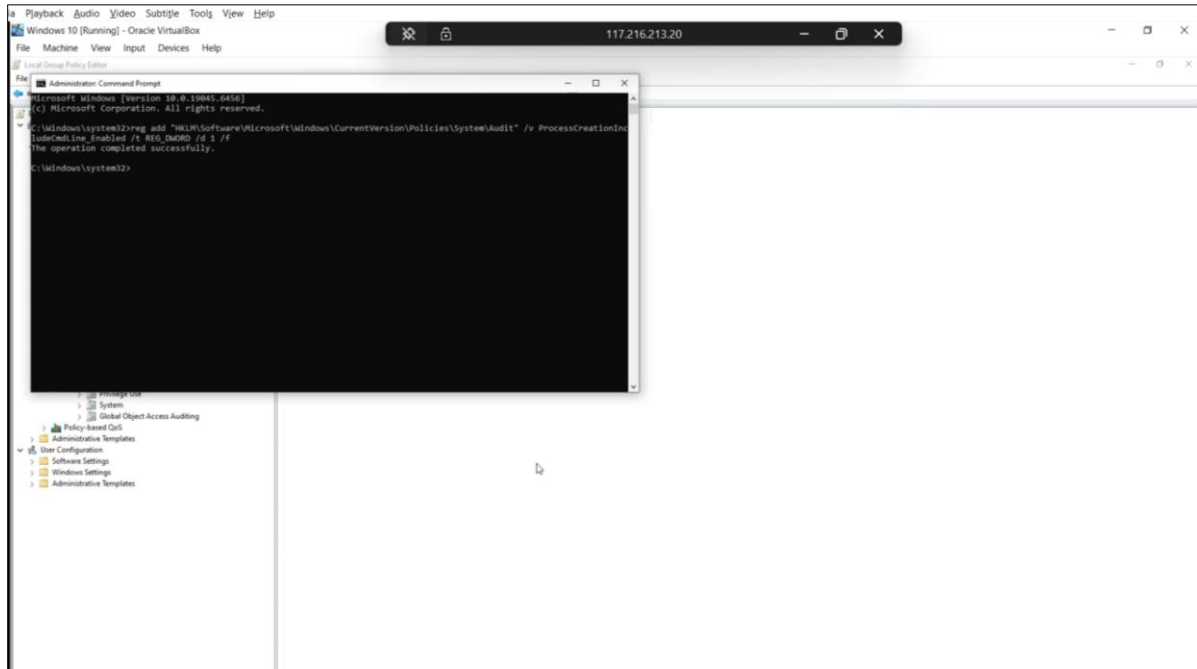
1. Open Command Prompt as Administrator

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

2. Run: reg add

“HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit
it” /v ProcessCreationIncludeCmdLine_Enabled /t REG_DWORD /d 1 /f



3.6 Wazuh Agent Log Forwarding Configuration

To enable log forwarding, the Wazuh agent configuration file (ossec.conf) was modified on the Windows 10 client.

Steps

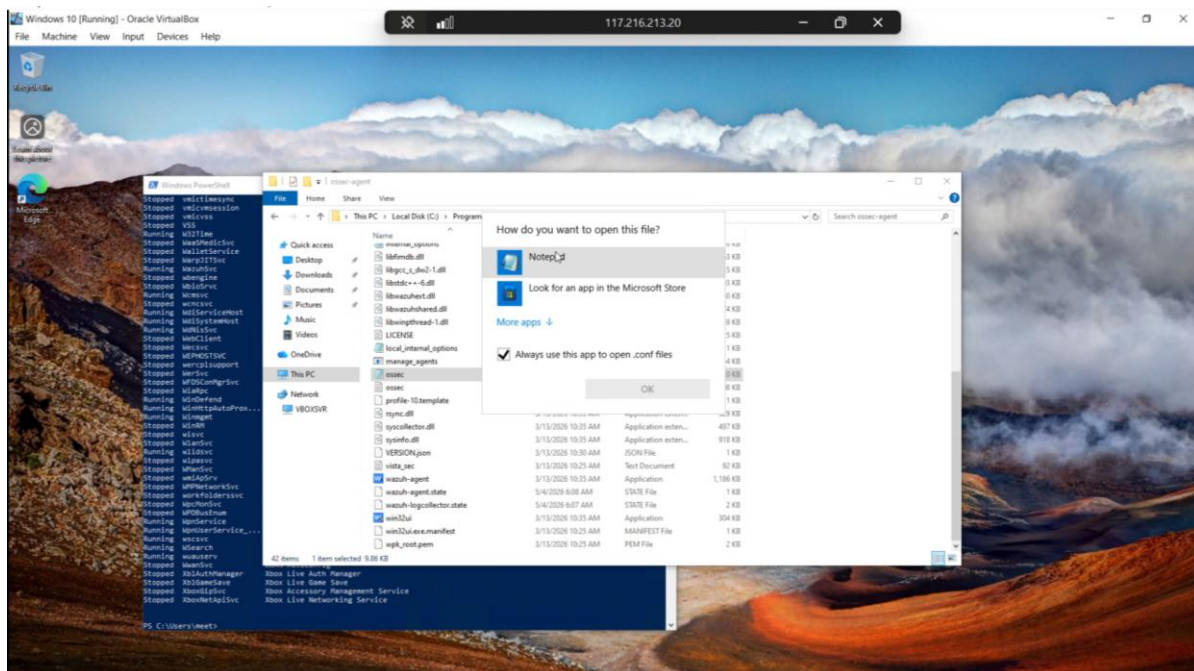
1. Open the configuration file: C:\Program Files (x86)\ossec-agent\ossec.conf
2. Add:

</Hacker4Help>

```
<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
</localfile>

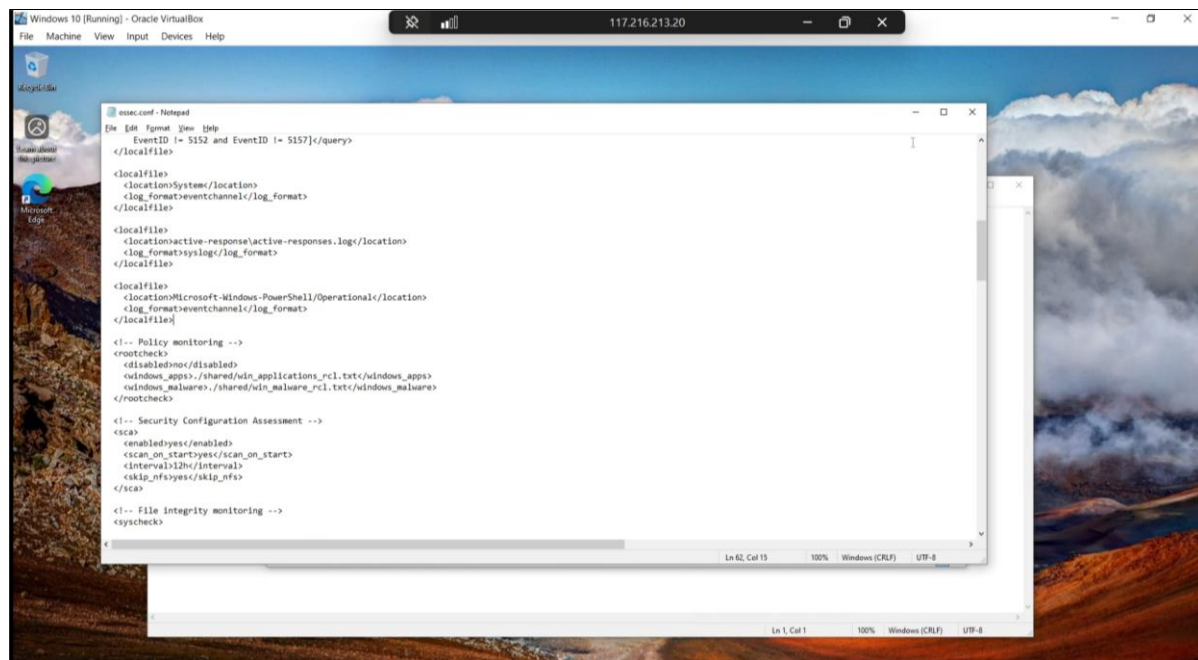
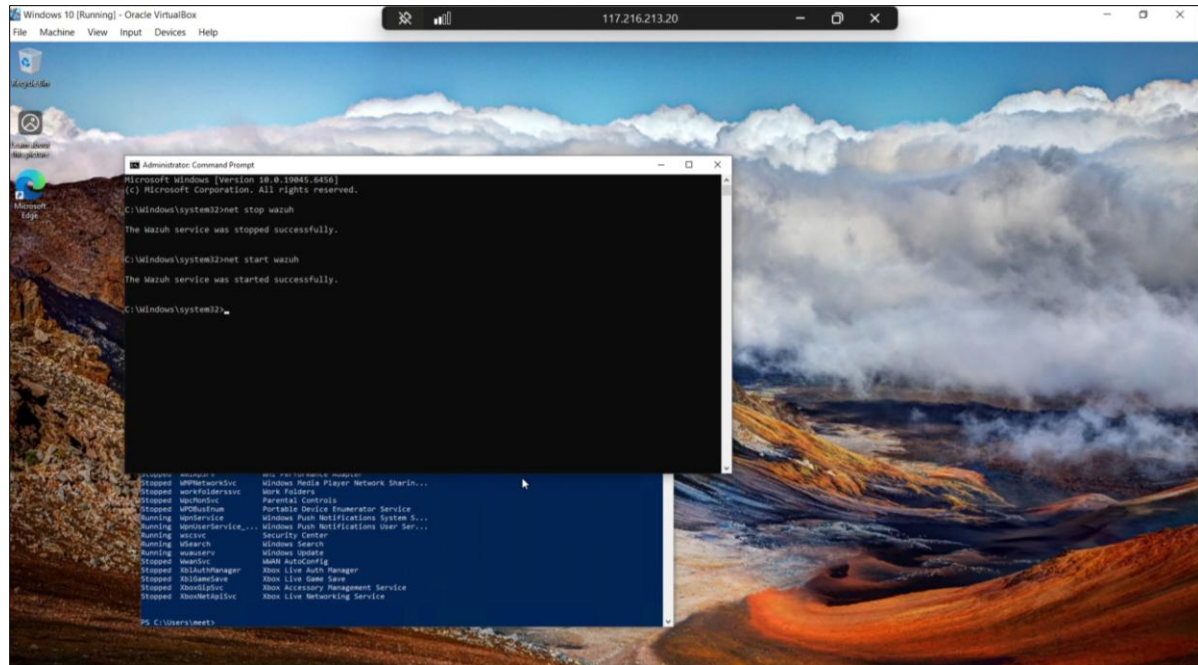
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

3. Open cmd and run “net stop wazuh” and “net start wazuh” to restart Wazuh agent



Ritik Patel
SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>



3.7 Outcome

After completing the configuration, the Windows 10 client successfully generated and forwarded detailed logs to Wazuh SIEM. This enabled full visibility into system

Ritik Patel
SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>

activity, forming the foundation for detecting and analyzing simulated attack scenarios.

4. Attack Scenario

To simulate common Windows operating system threats, a controlled attack scenario was designed within the lab environment. This scenario focuses on post-exploitation activities performed after an attacker has already gained access to the system.

4.1 Scenario Description

In this simulation, it is assumed that the attacker has successfully compromised the Windows 10 client machine and is now performing malicious actions to maintain access and avoid detection.

The attacker does not rely on external tools; instead, built-in Windows utilities such as PowerShell and command-line tools are used. This technique is commonly known as Living-Off-the-Land (LotL), where attackers abuse legitimate system tools to blend in with normal activity.

4.2 Target Details

- System: Windows 10 Client
- Role: Domain-joined endpoint
- Purpose: Execution of simulated attacker activities

4.3 Attacker Behavior

The attacker performs a sequence of actions to simulate real-world post-compromise activity.

The attack follows these steps:

1. The attacker executes a suspicious PowerShell command
2. A new user account is created on the system

</Hacker4Help>

3. The user account is modified
4. The attacker attempts to disable security controls
5. The attacker clears system logs to remove evidence

4.4 Techniques Used

The following techniques are simulated:

- PowerShell Abuse – Execution of suspicious commands
- Account Creation – Establishing persistence
- Account Modification – Changing system state
- Defense Evasion – Attempting to bypass security
- Log Clearing – Removing traces of activity

4.5 Attack Objective

The objective of the attacker is to:

- Maintain access to the compromised system
- Operate using legitimate tools to avoid detection
- Remove evidence of malicious activity

4.6 Expected Outcome

The attack is expected to generate multiple Windows event logs, including:

- PowerShell execution logs
- Account management logs
- Security log clearing events

These logs are then collected and analyzed using the Wazuh.

5. Attack Simulation

In this phase, multiple Windows operating system threats were simulated on the Windows 10 client machine. The objective was to replicate realistic attacker behavior and generate system logs that can be analyzed using the Wazuh.

5.1 Suspicious PowerShell Execution

Command Used:

IEX "Write-Host Attack_Simulation"

Explanation:

- IEX (Invoke-Expression) is used to execute commands dynamically
- This technique is commonly used by attackers in fileless malware
- It allows execution of scripts directly in memory

Attack Behavior:

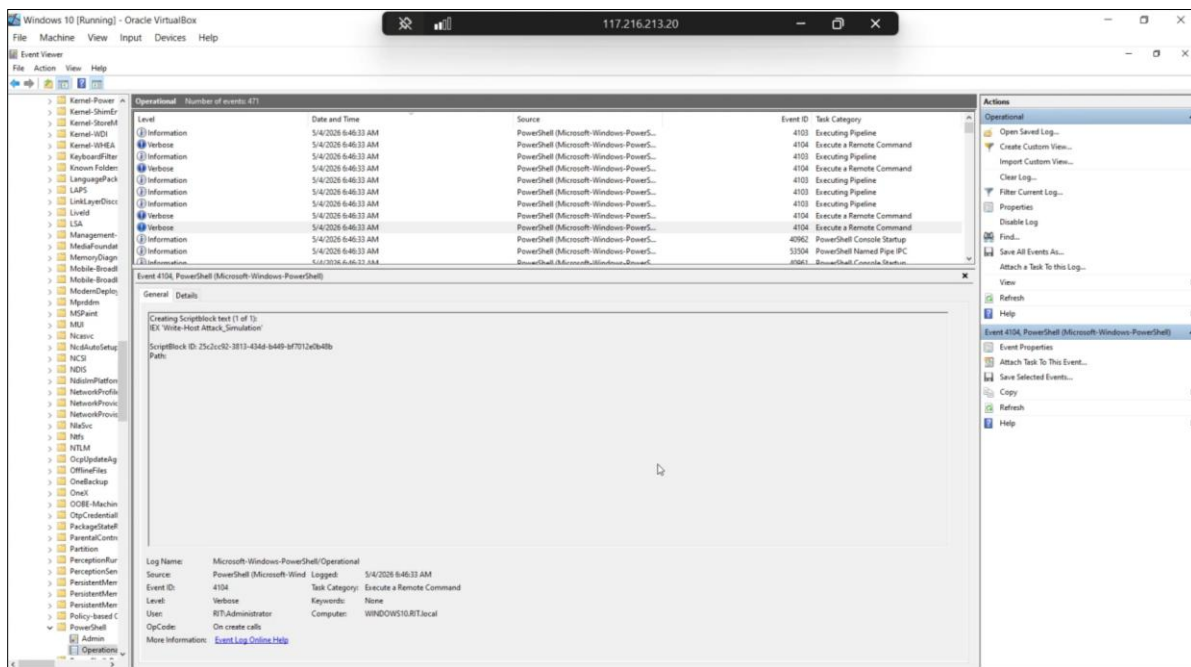
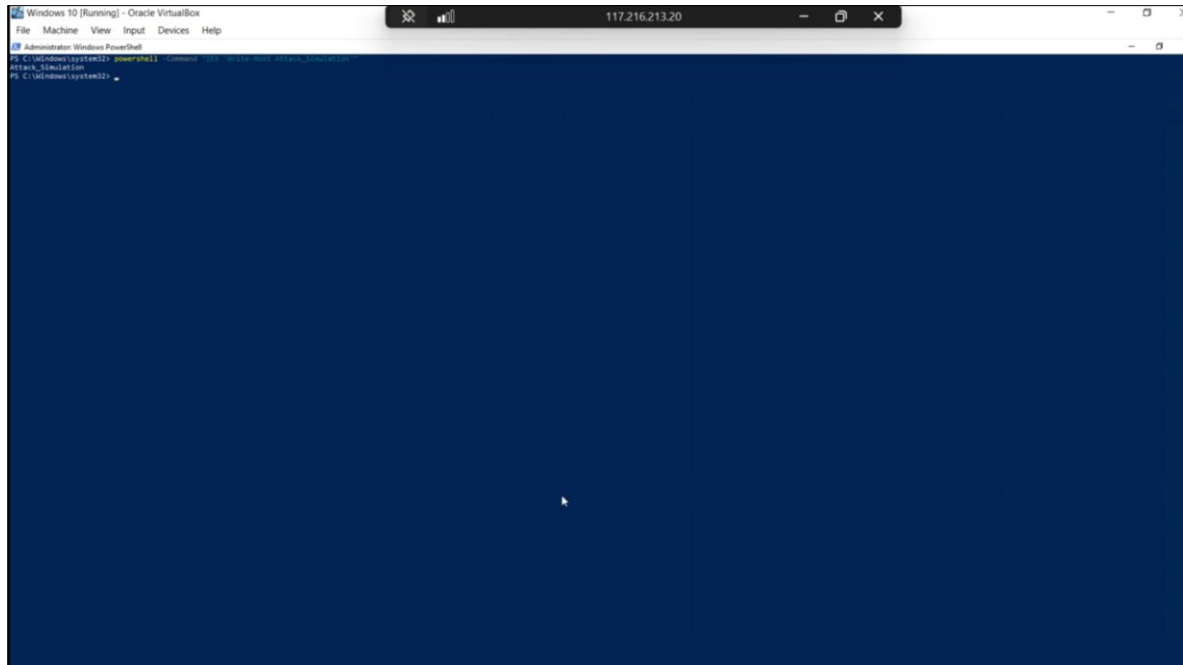
- A PowerShell command was executed on the system
- The command simulates malicious execution behavior
- No external file was required

Observation:

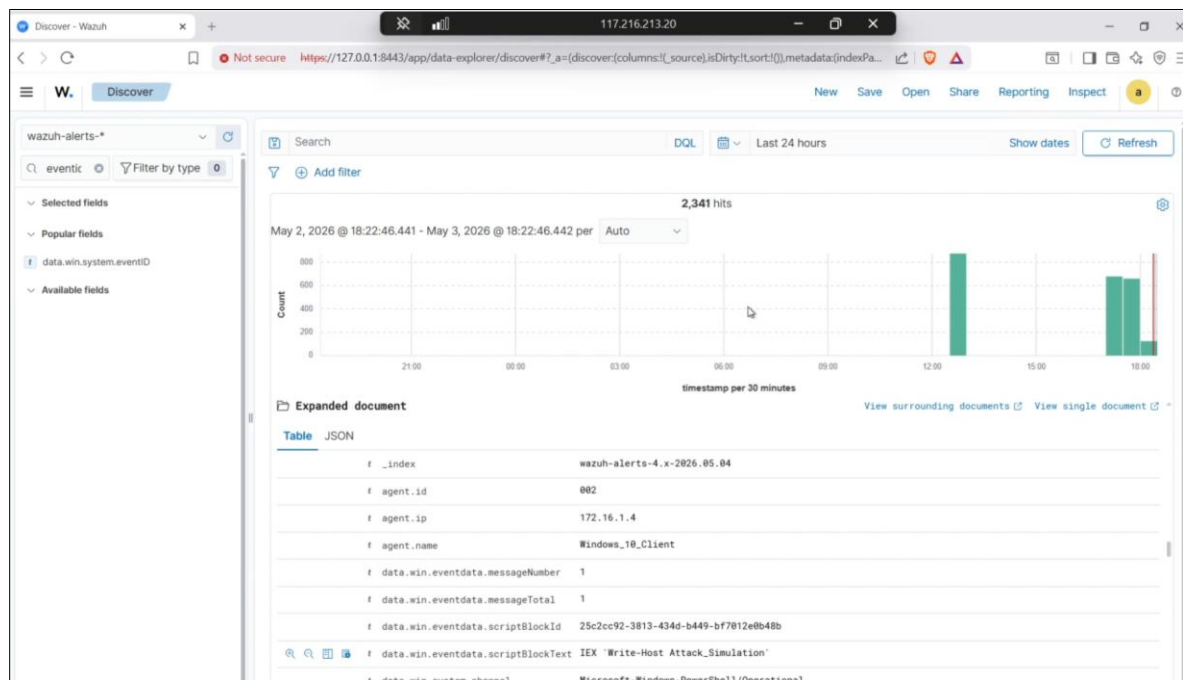
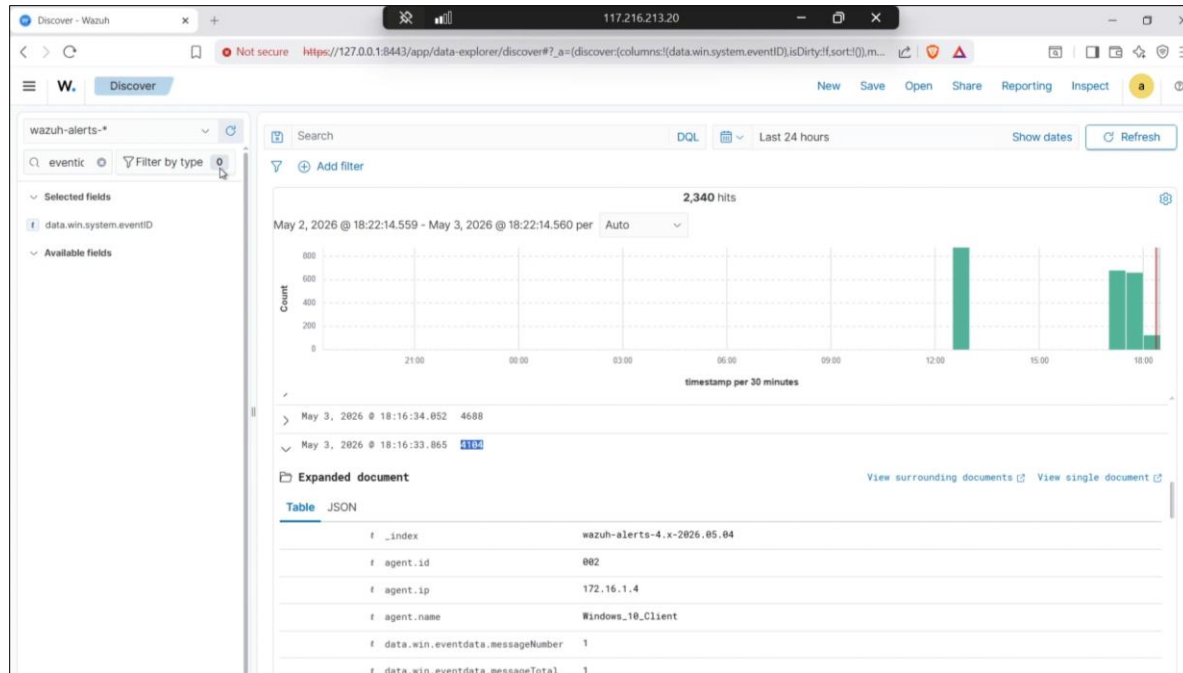
- Event ID **4104** was generated
- Script block logging captured the command
- The executed command was visible in Event Viewer

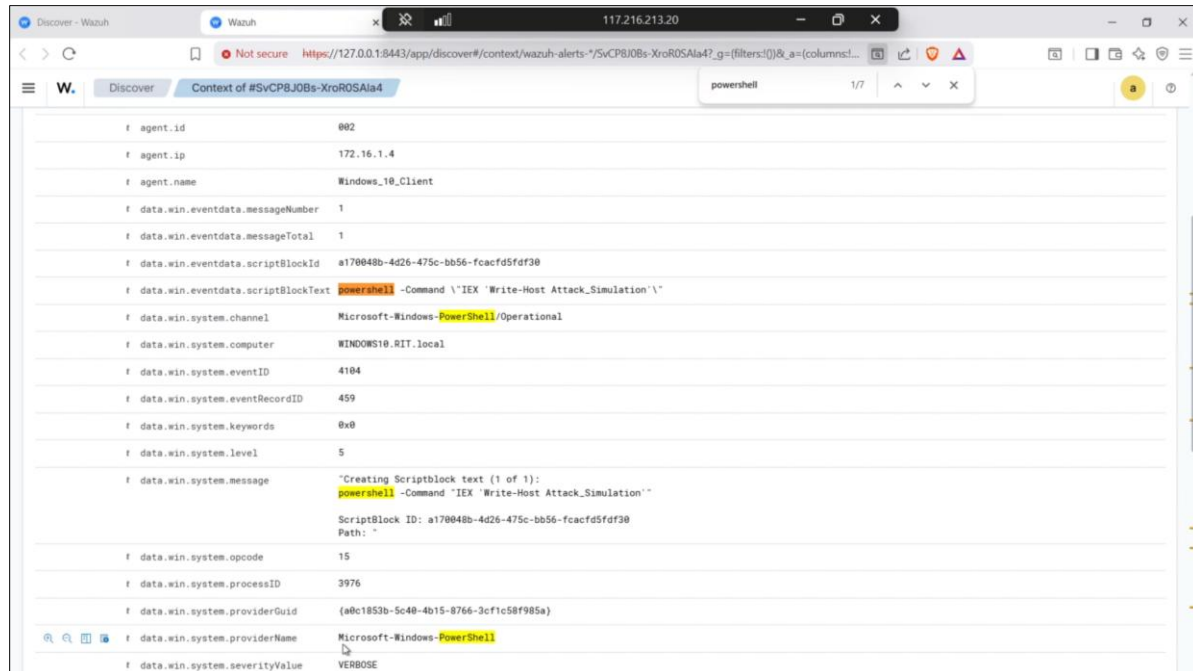
The use of IEX is suspicious because it allows dynamic execution of commands, commonly used in fileless malware and obfuscation techniques.

</Hacker4Help>



Ritik Patel
SOC - Threat Research intern at </Hacker4Help>





5.2 User Account Creation

Command Used:

net user attacker Ritik@Hacker4Help /add

Explanation:

- Creates a new local user account
- Used by attackers to maintain persistence

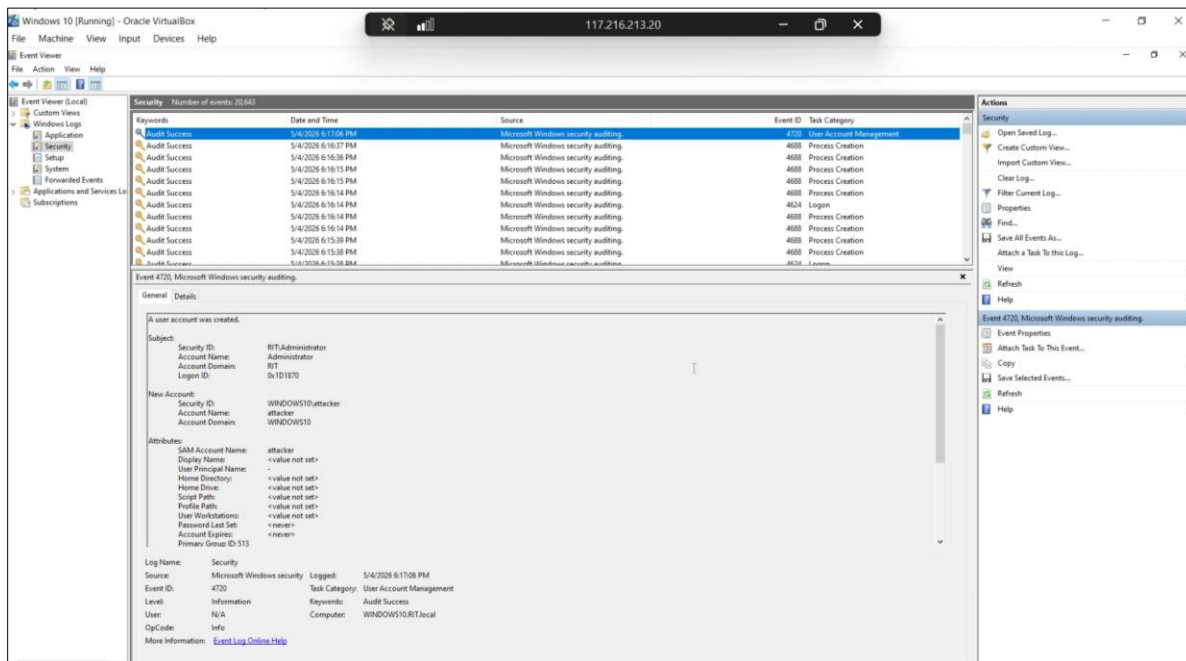
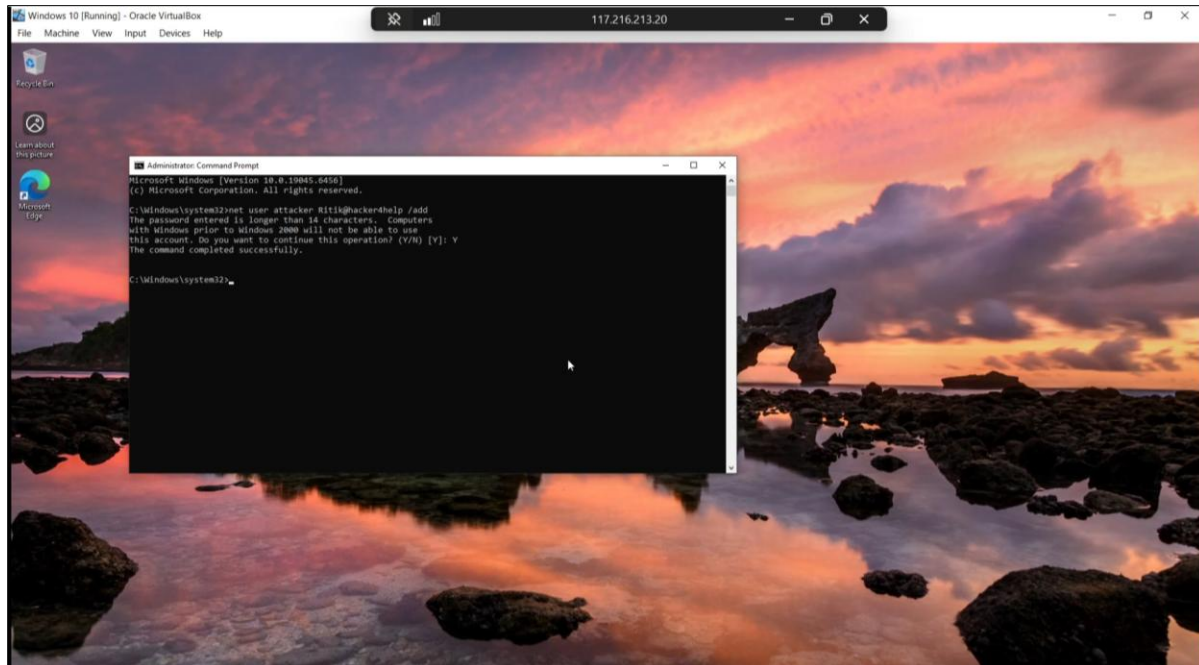
Attack Behavior:

- A new user account was created on the system
- This account can be used for future access

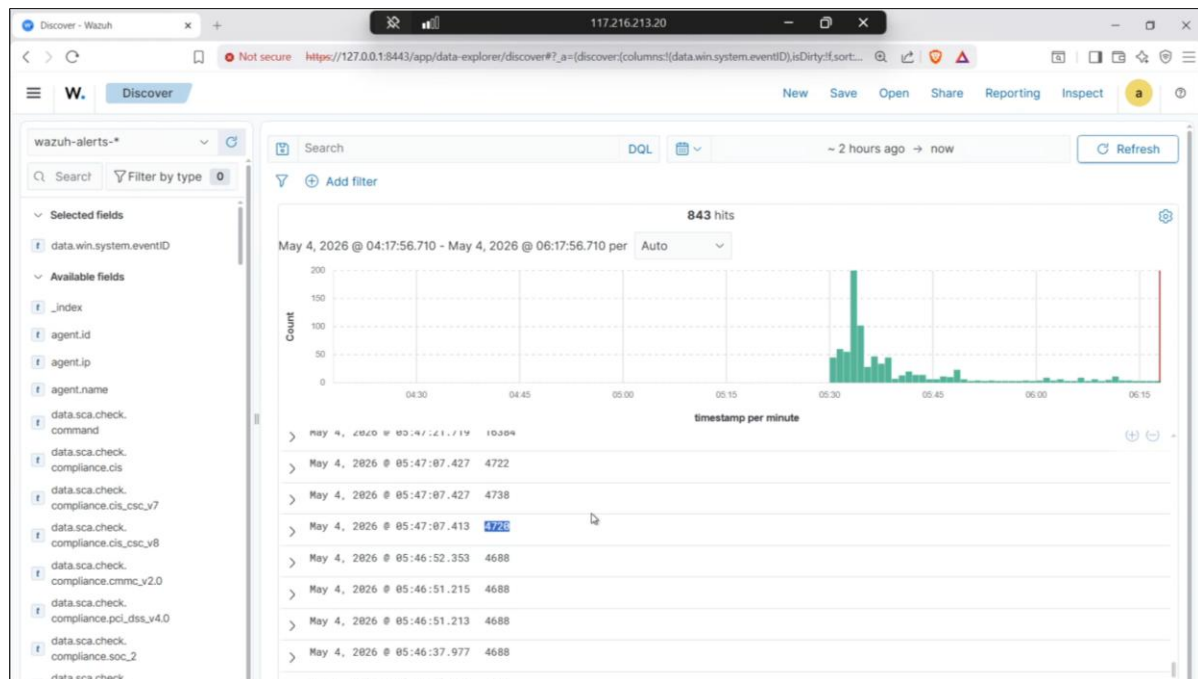
Observation:

- Event ID 4720 was generated
- The new user account was recorded in security logs

</Hacker4Help>



Ritik Patel
SOC - Threat Research intern at </Hacker4Help>



The screenshot shows the 'Context of #xXcH850BF2cAyLPePuO-' in the Wazuh Discover interface. It displays a table of fields and their values, including event data, system information, and a detailed message about a user account creation.

data.win.eventdata.targetSid	S-1-5-21-772672721-1922782846-2984232665-1082																
data.win.eventdata.targetUserName	attacker																
data.win.eventdata.userAccountControl	%%2080 %%2082 %%2084																
data.win.eventdata.userParameters	%%1793																
data.win.eventdata.userWorkstations	%%1793																
data.win.system.channel	Security																
data.win.system.computer	WINDOWS10_RIT.local																
data.win.system.eventID	4720																
data.win.system.eventRecordID	20597																
data.win.system.keywords	0x8020000000000000																
data.win.system.level	0																
data.win.system.message	A user account was created. Subject: <table><tr><td>Security ID:</td><td>S-1-5-21-397110547-3364043248-1258860236-500</td></tr><tr><td>Account Name:</td><td>Administrator</td></tr><tr><td>Account Domain:</td><td>RIT</td></tr><tr><td>Logon ID:</td><td>0x1D1870</td></tr></table> New Account: <table><tr><td>Security ID:</td><td>S-1-5-21-772672721-1922782846-2984232665-1082</td></tr><tr><td>Account Name:</td><td>attacker</td></tr><tr><td>Account Domain:</td><td>WINDOWS10</td></tr></table> Attributes: <table><tr><td>SAM Account Name:</td><td>attacker</td></tr></table>	Security ID:	S-1-5-21-397110547-3364043248-1258860236-500	Account Name:	Administrator	Account Domain:	RIT	Logon ID:	0x1D1870	Security ID:	S-1-5-21-772672721-1922782846-2984232665-1082	Account Name:	attacker	Account Domain:	WINDOWS10	SAM Account Name:	attacker
Security ID:	S-1-5-21-397110547-3364043248-1258860236-500																
Account Name:	Administrator																
Account Domain:	RIT																
Logon ID:	0x1D1870																
Security ID:	S-1-5-21-772672721-1922782846-2984232665-1082																
Account Name:	attacker																
Account Domain:	WINDOWS10																
SAM Account Name:	attacker																

5.3 User Account Modification

Command Used:

net localgroup administrators attacker /add

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>

Explanation:

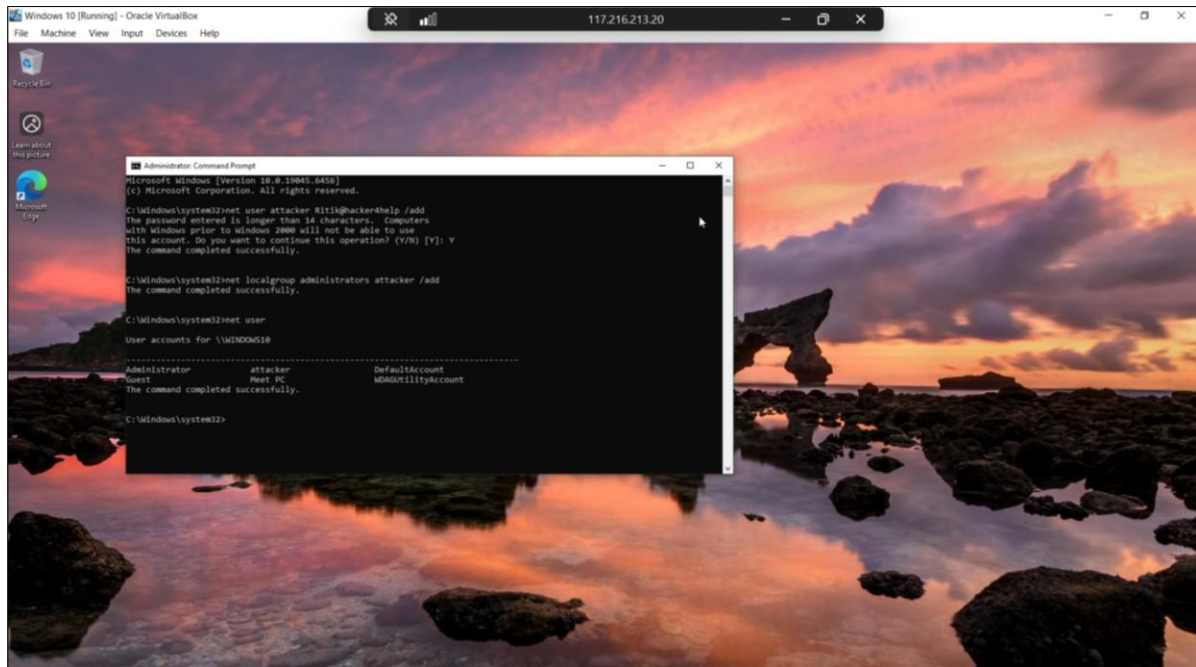
- Adding a user to a privileged group generates Event ID 4738 (Security Group Management), which is a stronger indicator of privilege escalation.

Attack Behavior:

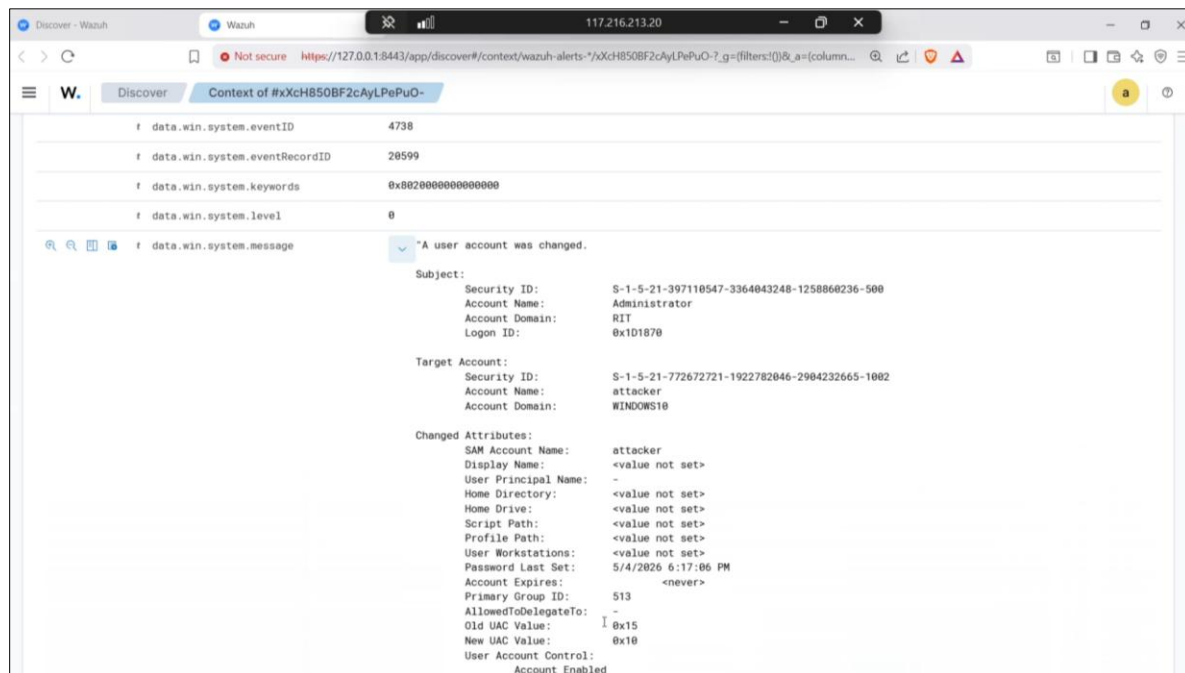
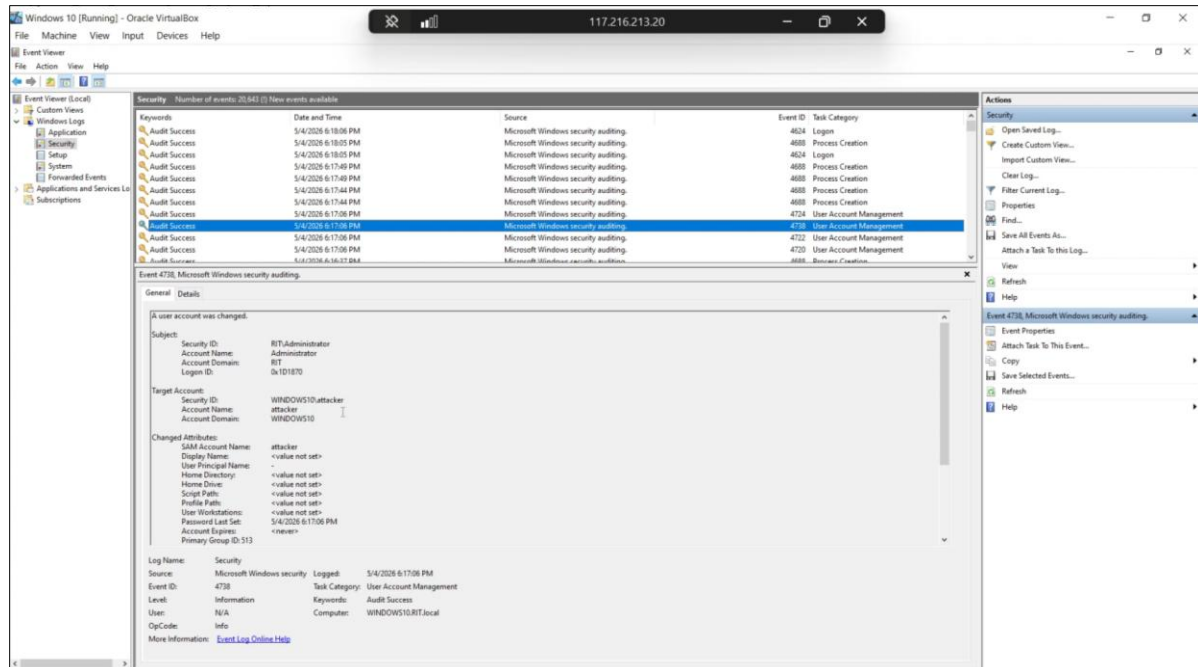
- The created user account was modified
- This indicates system-level changes

Observation:

- Event ID 4738 was generated
- Shows changes made to the user account



</Hacker4Help>



5.4 Defense Evasion Attempt (Windows Defender)

Command Used:

Set-MpPreference -DisableRealtimeMonitoring \$true

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>

Explanation:

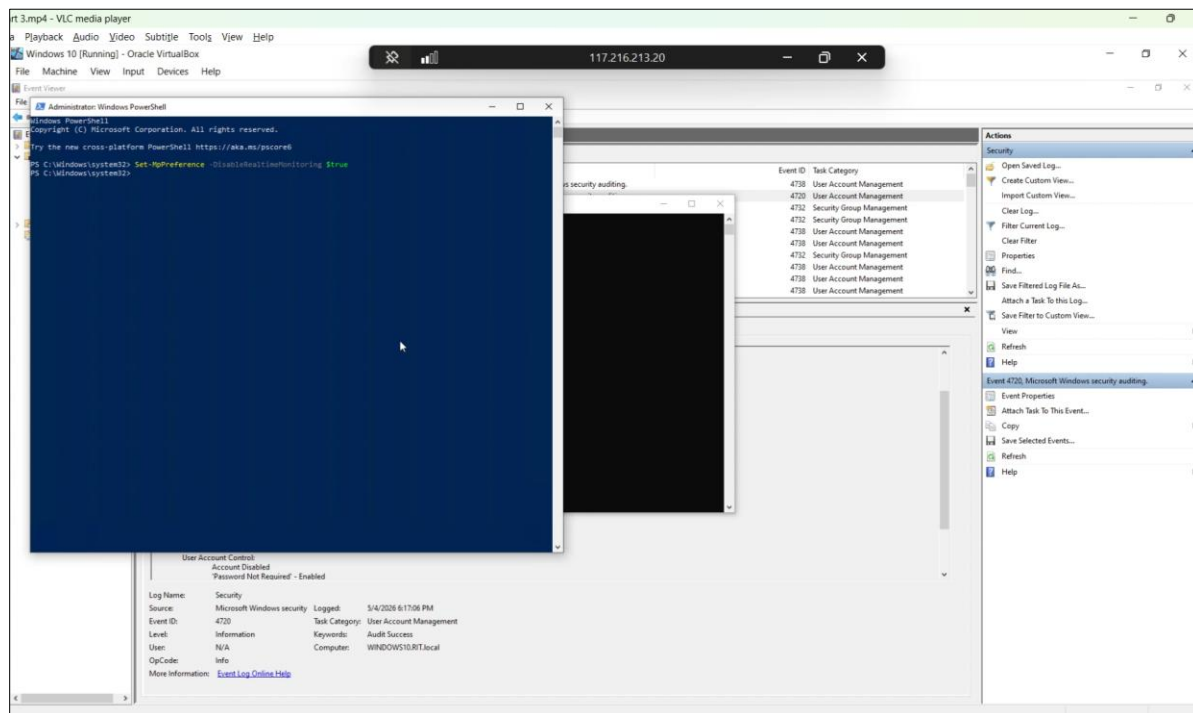
- Attempts to disable Windows Defender real-time protection
- Common attacker technique to avoid detection

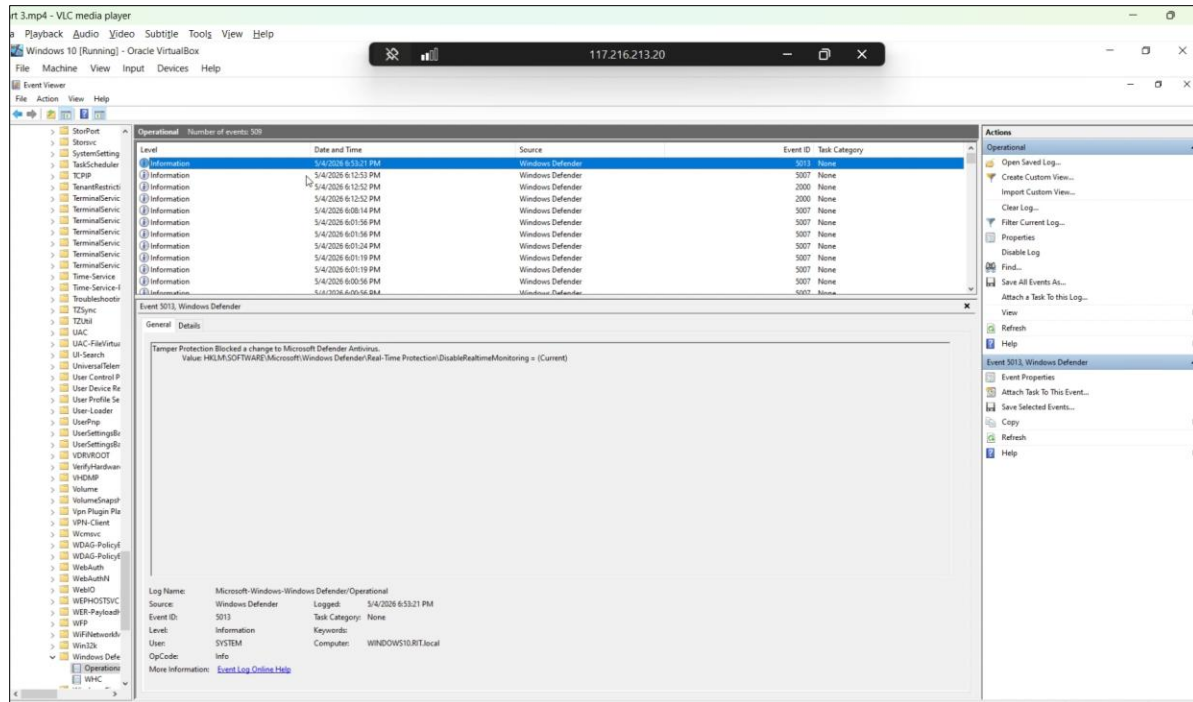
Attack Behavior:

- System attempted to modify Defender settings
- Tamper protection prevented the change

Observation:

- Event ID 5013 was generated
- Indicates security setting modification attempt





5.5 Clearing Security Logs

Command Used:

wevtutil cl Security

Explanation:

- Clears Windows Security logs
- Used by attackers to remove evidence

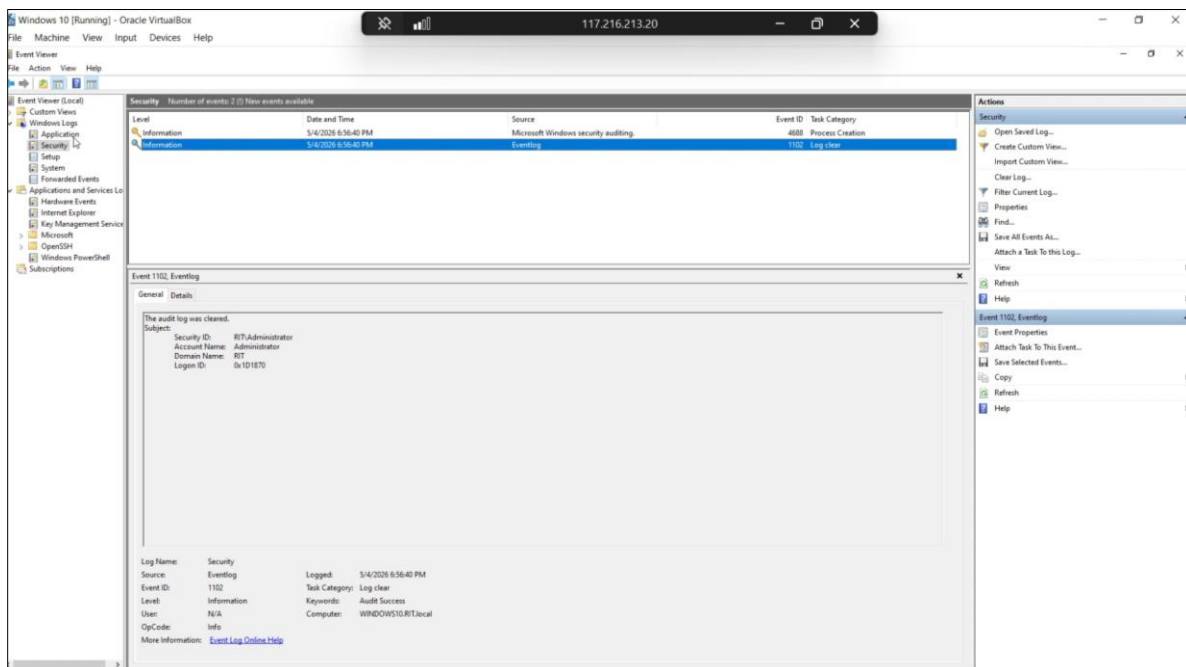
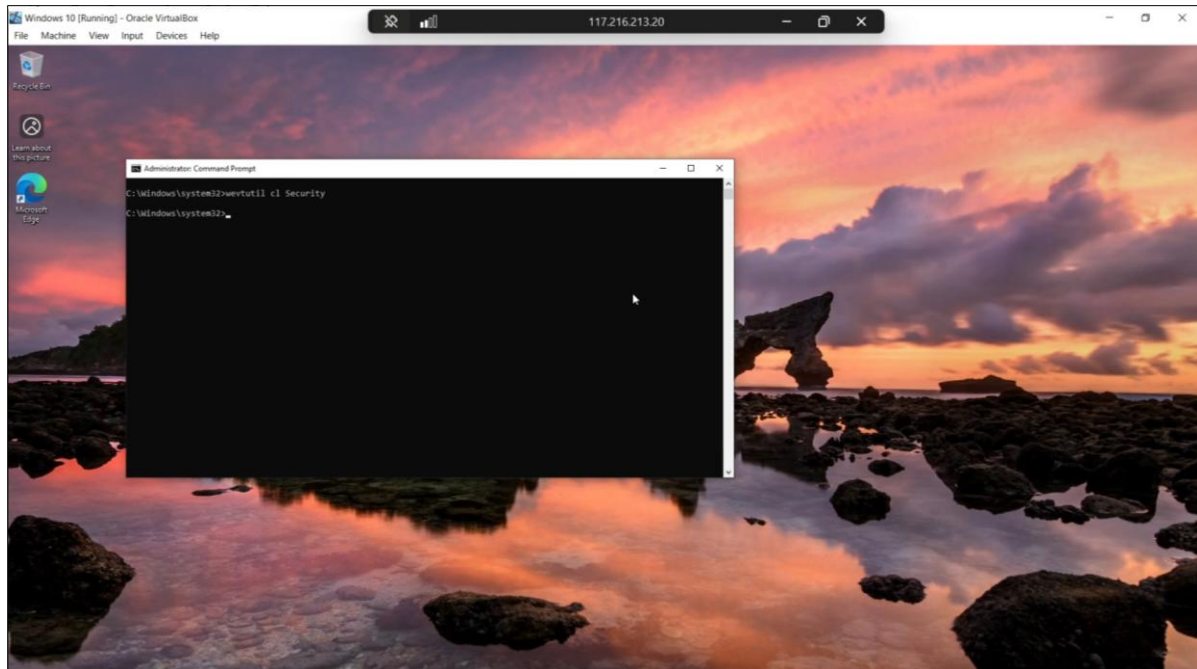
Attack Behavior:

- All security logs were cleared
- This is a strong indicator of malicious intent

Observation:

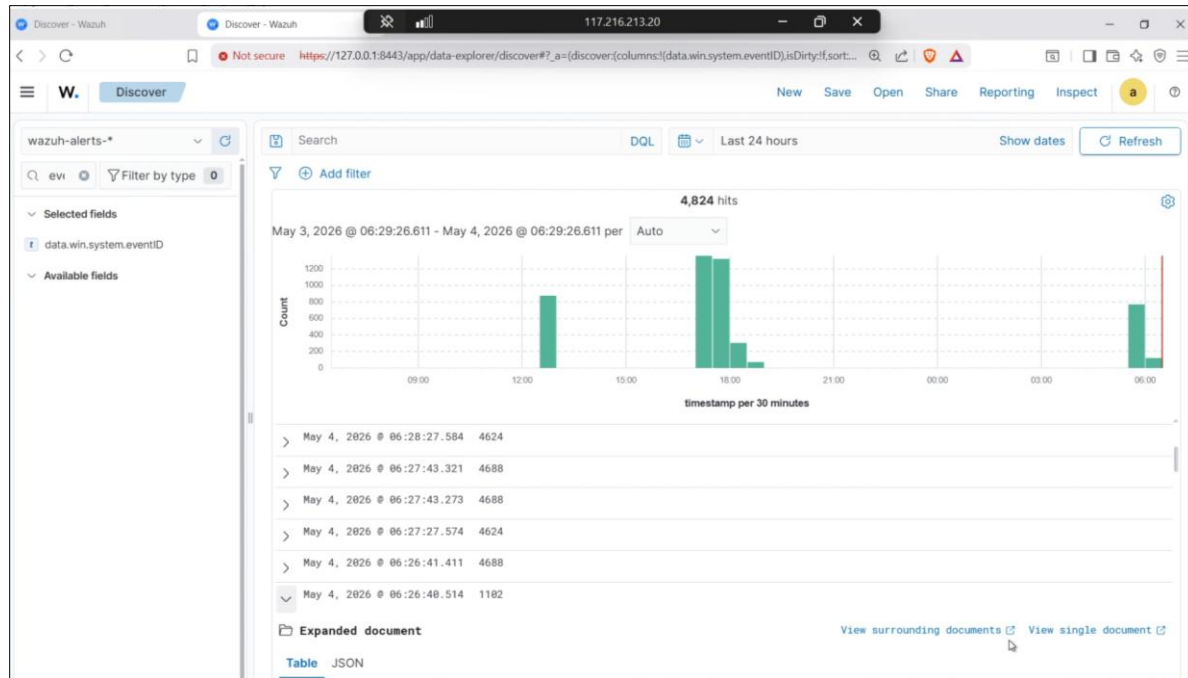
- Event ID 1102 was generated
- Indicates audit logs were cleared

</Hacker4Help>



Ritik Patel
SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>



The screenshot shows the Wazuh Discover interface with a specific document selected. The document details are as follows:

Field	Value
@timestamp	May 4, 2026 @ 06:26:40.514
_index	wazuh-alerts-4.x-2026.05.04
agent.id	902
agent.ip	172.16.1.4
agent.name	Windows_10_Client
data.win.logFileCleared.clientProcessId	1528
data.win.logFileCleared.clientProcessStartKey	5629499534213621
data.win.logFileCleared.subjectDomainName	RIT
data.win.logFileCleared.subjectLogonId	0x1d1870
data.win.logFileCleared.subjectUserName	Administrator
data.win.logFileCleared.subjectUserSid	S-1-5-21-397110547-3364843248-1258860236-500
data.win.system.channel	Security
data.win.system.computer	WINDOWS10.RIT.local
data.win.system.eventID	1102
data.win.system.eventRecordID	20731
data.win.system.keywords	0x4020000000000000
data.win.system.level	4
data.win.system.message	"The audit log was cleared."

Subject: Security ID: S-1-5-21-397110547-3364843248-1258860236-500
Account Name: Administrator

Ritik Patel
SOC - Threat Research intern at </Hacker4Help>

6. Detection And Analysis Using Wazuh

After executing the simulated attacks, the next step was to analyze how the activities were detected using the Wazuh.

Wazuh continuously collects logs from monitored endpoints through agents and applies detection rules to identify suspicious behavior. In this scenario, multiple events generated during the attack were captured and analyzed.

6.1 Identifying Relevant Security Events

The simulated attack generated multiple Windows event logs corresponding to different attacker activities.

Key Events Observed:

Event ID	Description
4104	PowerShell Execution
4720	User Account Created
4738	User Account Modified
1102	Security Log Cleared

Observation:

Each stage of the attack generated a corresponding event, allowing clear visibility into attacker actions.

6.2 Log Analysis in Wazuh Dashboard

To analyze the attack:

- Navigate to Wazuh Dashboard
- Go to Threat Hunting / Discover
- Apply filters for relevant Event IDs

6.3 Detection of Suspicious Activity

The logs indicate abnormal behavior based on:

- Execution of PowerShell using dynamic commands
- Creation of a new user account
- Modification of account attributes
- Clearing of security logs

Analysis:

- PowerShell Execution (4104) → Indicates suspicious command execution
- User Creation (4720) → Suggests persistence mechanism
- User Modification (4738) → Indicates system-level changes
- Log Clearing (1102) → Strong indicator of defense evasion

6.4 Attack Pattern Correlation

The sequence of events clearly shows a structured attack pattern:

1. PowerShell execution (initial activity)
2. User account creation (persistence)
3. Account modification (system manipulation)
4. Log clearing (defense evasion)

Key Insight:

These events are not isolated; when correlated together, they form a clear attack chain.

6.5 Wazuh Detection Mechanism

Wazuh processes logs as follows:

1. Events are generated on the Windows endpoint
2. Wazuh agent collects the logs

</Hacker4Help>

3. Logs are sent to the Wazuh manager
4. Detection rules are applied
5. Suspicious activity is identified

Alert Behavior:

- Single events → low context
- Multiple related events → strong indicator of compromise

6.6 MITRE ATT&CK Mapping

The simulated attack can be mapped to the MITRE ATT&CK framework:

Tactic	Technique
Execution	T1059 – Command and Scripting Interpreter (PowerShell)
Persistence	T1136 – Create Account
Defense Evasion	T1070 – Indicator Removal (Clear Logs)

6.7 Key Findings

- Suspicious PowerShell activity was successfully detected
- Unauthorized account creation was observed
- Account modification events were recorded
- Log clearing activity indicated defense evasion
- Multiple events were correlated to identify attack behavior

7. Conclusion

7.1 Summary

In this project, multiple Windows operating system threats were successfully simulated and analyzed within a controlled lab environment.

The setup included:

- Active Directory environment with a Windows Server acting as the Domain Controller
- Windows 10 client machine used as the target system
- Wazuh for centralized log collection and analysis
- A structured logging configuration to capture detailed system activity

The simulation focused on post-exploitation attacker behavior, where actions were performed after gaining access to the system. These actions included PowerShell execution, user account creation, account modification, and log clearing.

Each activity generated corresponding Windows event logs, which were successfully collected and analyzed using Wazuh.

7.2 Key Achievements

- ✓ Successfully simulated multiple Windows OS threats
- ✓ Generated relevant security events (4104, 4720, 4738, 1102)
- ✓ Enabled and configured advanced logging on Windows endpoint
- ✓ Detected and analyzed attacker behavior using Wazuh SIEM
- ✓ Correlated multiple events to identify attack patterns
- ✓ Demonstrated defense evasion techniques such as log clearing

7.3 Practical Learning Outcomes

This project provided hands-on experience with:

- Windows event logging and auditing
- PowerShell-based attack techniques
- SIEM-based monitoring and detection
- Log correlation and attack timeline analysis
- Post-exploitation attacker behavior

It also improved understanding of:

- How attackers operate after gaining access
- How legitimate tools can be abused for malicious purposes
- The importance of detailed logging in detection

7.4 Real-World Relevance

This simulation represents a realistic Security Operations Center (SOC) scenario where:

- An attacker gains access to an endpoint
- Performs actions to maintain persistence
- Attempts to evade detection by clearing logs
- Generates multiple correlated events

In real-world environments, such behavior is commonly observed during advanced attacks, where attackers rely on built-in tools and attempt to remain undetected.

This project demonstrates how SIEM solutions can be used to detect, analyze, and investigate such activities effectively.

7.5 Future Work

- Configure custom Wazuh detection rules for better alerting
- Enable additional log sources such as Windows Defender logs

Ritik Patel

SOC - Threat Research intern at </Hacker4Help>

</Hacker4Help>

- Implement alert correlation and automation
- Simulate additional attack techniques (e.g., lateral movement, privilege escalation)
- Integrate threat intelligence feeds